INCH-POUND

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31 October 1994
SUPERSEDING
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31 October 1989

MILITARY SPECIFICATION

TRUCK, LIFT, FORK, TELESCOPING MAST, ELECTRIC, SELF PROPELLED (WALKIE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for electric, industrial straddle or reach, stacker type forklift trucks. The rated load capacity is at 24 inches (610mm) from the front of the forks.

1.2 Classification. Unless otherwise specified (see 6.2), trucks shall be one of the following types and sizes:

Types:

Type E - Nonspark enclosed

Type EE - Spark enclosed

Beneficial comments (recommendations, additions, deletions) and any other pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Navy Ships Parts Control Center, Code 0341 P.O. Box 2020 , Mechanicsburg, PA 17055-0788 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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Sizes:

Size 1 - 1000 pounds (454 kilograms) capacity, 130 inches (3300 millimeters) lift, 83 inches (2108 millimeters) collapsed mast height

Size 2 - 2000 pounds (908 kilograms) capacity, 130 inches (3300 millimeters) lift, 83 inches (2108 millimeters) collapsed mast height

Size 3 - 3000 pounds (1362 kilograms) capacity, 130 inches (3300 millimeters) lift, 83 inches (2108 millimeters) collapsed mast height

Size 4 - Capacity, Lift Height, and Collapsed Mast Height as specified (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation, (see 6.2).

SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photosensitive Annodized Aluminum)
QQ-C-320	- Chromium Plating (Electrodeposited)
QQ-P-416	- Plating, Cadmium (Electrodeposited)
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant
W-B-133	- Batteries, Storage (Lead-Acid, Industrial for Portable Service)

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MILITARY

MIL-P-514 - Plates, Identification, Instruction, and Marking, Blank

MIL-F-3541 - Fittings, Lubrication

MIL-M-3971 - Meter, Time Totalizing, Non Hermetically Sealed, Electric, General Specification for

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine Combat/Tactical Service for

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

FED-STD-595 - Colors

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-162 - Materials Handling Equipment, Preparation for Shipment,Storage, Cyclic Maintenance, Routine Testing and Processing

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099

2.1.2 Other Government documents, drawings and publications.The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF LABOR

29 CFR 1910-Occupational Safety and Health Standard (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein.

Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B56.1 - Safety Standards for High Lift and Low Lift Industrial Trucks

B56.11.3 - Load Handling Symbols for Powered Industrial Trucks

(Application for copies should be addressed to: American National Standards Institute,1430 Broadway, New York, New York 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

ASTM D 3951- Packaging, Commercial

(Application for copies should be addressed to: American Society for Testing and Materials, 1961 Race Street, Philadelphia, PA 19103)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J514 - Hydraulic Tube Fitting

SAE J516 - Hydraulic Hose Fittings

SAE J517 - Hydraulic Hose

SAE J518 - Hydraulic Flanged Tube, Pipe and Hose Connections, 4-Bolt Split Flange Type

SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Device (30-1000 Mhz)

(Application for copies should be addressed to: Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

UNDERWRITERS' LABORATORIES INC.

UL 583- Standards for Safety for Electric Battery Powered Industrial Trucks

(Application for copies should be addressed to: Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, New York 11746; 333 Pfingston Road, Northbrook, IL 60062 or 1655 Scott Boulevard, Santa Clara, California 95050).

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the reference cited herein(except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this document supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a first produced unit shall be subjected to a first article inspection (see 6.2) in accordance with 4.4.

3.2 General. Trucks covered by this specification shall be complete and ready for operation when delivered. Trucks shall be constructed and equipped to meet all requirements specified herein.

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The end item and major and minor assemblies shall be new and of a standard design or model product of the truck manufacturer current at the time of delivery, except for such modifications from manufacturer's standard as may be required to conform to these specification requirements. The component parts of the unit need not be the products of the same manufacturer. However, the manufacturers shall be established in the industry so that prompt and continuing service and delivery of repair parts will be assured.

3.3 Material. Material shall be as specified herein. Materials not definitely specified shall be equal to the end item manufacturer's normal commercial product and shall meet the test requirements specified in section 4.

3.4 Design. Design of trucks covered by this specification and all components incorporated therein, shall conform to accepted design practices and shall insure safe, efficient and economical operation under normal service conditions. Trucks shall be powered by a 24-volt industrial battery which shall energize and operate two electrical d.c. motors on separate electrical circuits. One motor shall power the drive wheel(s) through a gear reduction unit. The second motor shall drive a hydraulic pump to provide power for lifting and tilting operations. Trucks shall be steered manually by means of a steering handle through the drive wheel(s). Controls for forward and reverse travel and speed changes shall be located on the steering handle for easy access of the operator.

3.4.1 Safety. Trucks shall conform to the applicable requirements of UL 583 and OSHA 29 CFR, Chapter XVII, Part 1910. Trucks shall bear either the Underwriters' Laboratories or an independent testing laboratory label indicating their stamp of approval for the safety rating. All rotating and reciprocating parts of the forklift, excluding the mast, shall be enclosed or guarded.

3.4.1.1 Certificates. The contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the trucks furnished under this specification meet or exceed the requirements of UL 583 Type E or Type EE as specified. Acceptable evidence of meeting these requirements shall be certified test reports from recognized independent testing laboratories acceptable to the government, indicating that the trucks conform to the requirments of UL 583.

3.4.2 Bearings. Unless otherwise specified herein, rotating parts shall be mounted on ball or roller bearings.

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3.4.3 Lubrication. Accessible means shall be provided for applying and, where necessary, checking the level of lubricant in all parts and components. Oil drain plugs and connections shall provide for instant drainage using ordinary maintenance shop tools. Grease fittings shall be provided where required and shall conform to MIL-F-3541.

3.4.4 Maintainability. Provision shall be made for adjustment, servicing and replacement of all electrical assemblies and components, battery, wearing parts of lift and tilt mechanism, brake and components wearing parts of steering assembly, tires, wheels and lights. If hand access openings are used on the exterior of the truck, the opening shall be provided with a removable or hinged cover. Covers shall be secured with bolts or quick opening fasteners removable with ordinary hand tools. There shall be no interference to the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions.

3.4.5 Screw threads. Screw threads shall be in accordance with FED STD H28.

3.4.6 Workmanship. The truck shall perform any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by improper workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist. Pivot points shall have hardened pins and bushed holes.

3.4.6.1 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately punched or drilled and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, and in full contact with the surface of the member. Excessive upsetting of rivets to fill holes will not be acceptable.

3.4.6.2 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures and rough projecting edges and shall be thoroughly fused to the base metal.

3.4.6.3 Bolt connections. Bolt holes shall be accurately formed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs, and cap screws having straight threads. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers.

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All fasteners shall be properly tightened and nuts shall have full thread engagement.

3.4.6.4 Castings. Castings shall be free from blowholes, porosity, shrinkage, cracks, and other defects. Castings shall be commercially machinable and shall not be repaired, plugged or welded to adversely affect the strength, use, or machinability. Sufficient finish shall be allowed to permit machining for leak-free sealing at covers, closure points and fasteners.

3.4.6.5 Machine work. All parts shall be manufactured to insure accuracy of assembly through the use of correct jigs, fixtures or tape controlled machines or any combination of these. Like parts shall be interchangeable.

3.4.6.6 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having eroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide for interchangeability of components.

3.5 Electrical equipment. All electrical equipment and wiring incorporated in the trucks shall conform to the applicable requirements of Publication UL 583, unless otherwise specified herein.

3.5.1 Resistors. Resistors shall be of the heavy duty type rated for continuous use and shall not result in burn-out or fusion of resistors when tested in accordance with section 4. For type EE trucks, the use of resistors shall be avoided to the maximum extent practicable.

3.5.2 Conductors. Identification of circuits shall be provided through color coding, numbered markings or similar means to identify conductor runs. For type EE trucks, all conductors, except battery leads, emerging from enclosures shall pass through grommet type fittings so that sparks shall not propogate into the atmosphere and dust from the atmosphere shall not enter into the enclosure.

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3.5.3 Overcurrent devices. Overcurrent devices shall be provided on each power circuit. These devices shall be sized to carry the rated current of the battery at the one-hour rate or at maximum current drain when traveling on a level surface and lifting rated load, whichever is greater. In addition, an overcurrent device of duplicate size shall be provided for the control circuit network.

3.5.4 Enclosures and shields - type EE. Type EE electric trucks shall comply with the requirements of UL 583 except that no external surface or normally exposed part of the truck shall exceed 250 degrees Fahrenheit (120 degrees C) when subjected to any test specified herein. All electrical components including wiring shall be positioned or adequately protected from contact with excess moisture or battery electrolyte in the event of spillage during normal battery exchange.

3.5.5 Resistor housings - type EE. Resistor housings having horizontal surfaces which might accumulate combustible particles shall be avoided to the maximum extent practicable. The outside surface temperature of resistor housing shall stabilize below 250 degrees F (120 degrees C) or 150 degrees F (65 degrees C) above ambient, whichever is less when truck is operated continuously in low speed with rated load. The resistor housing outside surface may be cooled by oil coolant, fins, or other mechanical means to aid in dissipation of heat. When oil is used, there shall be no leakage from oil container, and oil shall not contact control resistors.

3.5.5.1 Thermal relays - type EE. Not less than two safety thermal relays shall be mounted on the resistor housing outside surface at critical heat areas as determined by the test specified in 4.5.2.15. The relays shall be preset to instantaneously disconnect the battery power to the affected circuit(s) if the resistor housing outside surface temperature reaches 250 degrees F (120 degrees C).

3.5.5.2 Static discharge device - type EE. Type EE trucks shall be equipped with static discharge devices in accordance with UL 583.

3.6 Motors and motor accessories.

3.6.1 Motors - Type E trucks. Motors shall be series or compound wound, heavy duty industrial type with a high starting torque designed for most efficient operation at 24 volts. Motors shall be capable of withstanding the tests specified in section 4 without evidence of damage or failure.

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3.6.2 Motors - Type EE trucks. In addition to the requirements of 3.6.1, drive motor(s) for type EE trucks shall be equipped with one or more thermal overload relays to open the motor power circuit if the temperature of the drive motor housing outside surface reaches 250 degrees F (120 degrees C) or 150 degrees F (65 degrees C) above ambient, whichever is less when truck is operating continuously in low speed with rated load. This requirement shall be determined by tests specified in 4.5.2.15 and 4.5.2.16.

3.6.3 Insulation. Insulation shall be selected to give the best balance of over-all operating characteristics of torque, speed, and temperature rise. The class of insulation shall be determined by the material with the lowest temperature rise used in the insulation system. Armature and fields need not have the same class of insulation, but each part shall meet the temperature rise of its class of insulation. The electrical and mechanical properties of the insulated winding shall not be impaired by the application of the temperature permitted for the insulating material. The word "impaired" is used in the sense of causing any change which could disqualify the insulating material for continuous service.

3.6.4 Speed Controller. The speed controller shall provide not less than two forward and two reverse speeds. The controller shall: (1) provide means to permit precision maneuvering in restricted areas, (2) provide smooth acceleration between each successive speed range, and (3) have automatic electrical or mechanical interlocks to prevent starting or reversing except through forward or reverse first speed, respectively. There shall be maximum interchangeability of parts between contactors on a given truck. Contactors shall operate on 24 volts dc and shall be replaceable. There shall be no evidence of failure in any part of the controller when the truck is tested as specified in section 4. When specified, (see 6.2), the speed controller shall be the solid state type and shall provide stepless speed control in both forward and reverse directions.

3.6.5 Battery and battery accessories.

3.6.5.1 Battery. Unless otherwise specified (see 6.2), batteries will be furnished by the supplier. Batteries shall be 24 volts, industrial, rated at not less than 220 ampere hours capacity for size 1 truck, 300 ampere hours capacity for size 2 truck, 340 ampere hours capacity for size 3 truck and as specified for size 4 truck (see 6.2) at the 6-hour rate and shall meet the requirements of W-B-133. Battery leads shall be equipped with type EC or SB connectors.

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3.6.5.2 Battery connectors. Unless otherwise specified (See 6.2) trucks shall be equipped with type EC mounting half connectors or SB type connectors, suitably located to fit the half specified in 3.6.5.1.

3.6.5.3 Battery well. The battery well shall be constructed to accommodate the battery specified in 3.6.5.1. The inside of well shall be protected with acid-resistant paint conforming to TT-C-494, type III.

3.6.6 Drive unit. All rotating parts shall be supported on anti-friction bearings. The drive shall be a gear reduction unit operating in a constant oil bath. Gear housing shall be equipped with a filler plug and a magnetic drain plug. The motor reduction unit and drive wheel assembly shall be integrally mounted and shall revolve on anti-friction bearings. With the drive wheels locked, the drive assembly shall be capable of withstanding a torque applied to the output of the drive motor equal to 150 percent of the stall torque (as published by the motor manufacturer) without failure, fracture, or permanent deformation of any component. When test is required (see 6.2), the complete drive assembly may be tested before or after complete assembly of the truck at the option of the contractor.

3.6.7 Hydraulic system. Power used for lifting, tilting, and other auxiliary hydraulic devices shall be furnished by a hydraulic pump(s) directly driven by an electric motor(s) independent of the traction motor(s). The lift system shall provide for smooth lowering of load at a speed not greater than 80 feet per minute (25 meters per minute) in the event of failure or damage to hydraulic lines. Provision shall be made to protect the hydraulic system against excessive pressures caused by overloading. The hydraulic reservoir and system shall be capable of being drained without spillage. The sump tank shall be equipped with baffles. It shall have an access for manual cleaning of the entire inside or be removable for cleaning. The sump tank breather and filler cap and the lift, tilt, and sideshift (when furnished) cylinders shall be provided with means to prevent contamination of hydraulic fluid by direct impingement or seepage of water. A removable screen which provides at least 50-mesh protection shall be installed in the suction line between the hydraulic reservoir and the pump. A full flow in-line filter 25-micron absolute or 10-micron absolute or 10-micron with 98-percent efficiency on the first pass, and with emergency bypass, shall be in the suction, the pressure, or the return line and shall be located for servicing without the necessity of entering the sump tank. When the in-line filter is installed in the suction line, its relative position with respect to the 50-mesh strainer is optional, but it shall not bypass the 50-mesh strainer.

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The design and location of the filters shall be compatible with the pump and other components of the hydraulic system. The in-line filter shall include an indicator to show when maintenance is required and shall require not longer than 10 minutes for changing the filter element. A 10-micron breather cap shall be provided. The sump tank shall be provided with a dipstick or other means for checking the fluid level, and means shall be provided to secure the dipstick, when furnished in place. The level indicator shall be graduated to show the amount required for filling, and shall be accessible without the use of handtools. The sump tank shall be of welded steel construction, be initially clean, and free of metal particles and other foreign material. The hydraulic system shall be designed to insure that the maximum hydraulic oil temperature does not exceed 250 degrees F (121 degrees C) on EE trucks under test conditions of Section 4. For Type EE trucks, two or more 250 degrees F (121 degrees C) thermal relays shall be installed in different critical heat areas on the same hydraulic reservoir and each one, separately, shall cause disconnection of power to the pump motor(s) before the temperature reaches 250 degrees F (121 degrees C) as applicable. The tank shall have a capacity not less than 110 percent of the amount of fluid required for the hydraulic system's operation; it shall have sufficient capacity to prevent air from entering the hydraulic system with all hydraulic pistons fully extended and shall have sufficient free air capacity to prevent oil from being discharged through the tank from the system. Lifting and other cylinders, if any, shall be provided with "wiper rings" to preclude entry of dust or dirt into the packing. Hydraulic valves and cylinders shall have means for preventing oil from dripping or running onto electrical elements. Fluid in the hydraulic system shall conform to MIL-L-2104 grade 10, or shall be commercial hydraulic fluid with rust and oxidation inhibitor, anti-foaming, anti-wear, turbine grade base, and initially not higher in contamination level than will pass through the in-line filter specified. Hydraulic cylinders which employ pivot type mountings shall be provided with means for lubrication of the bearing area at such pivots. The use of self lubricated bearings is permissible. A removable strainer on the suction side of the reservoir and a combination dipstick/breather cap is acceptable.

3.6.7.1 Hydraulic fitting, hoses, and tubing. All components utilizing hydraulic fluid from the hydraulic reservoir excluding brakes and steering systems shall conform to the following standards. Hydraulic fittings used on valves and cylinders shall be the "O" ring straight thread type in accordance with SAE J514 but not limited to the fittings listed therein. When hydraulic tubing is used, fittings shall conform to SAE J514 37 degrees two- or three-piece tube assembly.

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All other fittings shall be of the 4-bolt split flange type in accordance with SAE J518 or the "O" ring straight thread type except for barbed or clamp types at both ends of suction, return and cylinder bleeder lines, and dry seal connectors on lines where hydraulic pressure does not exceed 200 psi (1300 KPa). Hydraulic hose specification with respect to applications shall conform to the following:

APPLICATION	HOSE SPECIFICATION
Suction and discharge	SAE J517,100R3 or 100R4
Pressure	SAE J517 100R1, 100R2 100R5 or 100R7

3.6.7.2 Hose reels. When hydraulic hose reels or other types of takeups are used on the mast to contain hydraulic hose, they shall be designed to facilitate containment, release, and retraction of the hose without entanglement. Sheave wheels, guide rollers, guide blocks, or a mechanism in the hose reel may be used to maintain hose tension.

3.6.7.3 Piston rods. Piston rods for the hydraulic system shall be plated with an electro-deposited hard chrome-plate finish conforming to type I, Class 2 of QQ-C-320. The chrome-plate finish shall be not less than 0.0005 inch (.01 mm) thick.

3.7 Structure.

3.7.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel and steel castings equal to the size and design of the manufacturer's commercial frame members. Frame members, bracings, and all joints shall provide a rigid unit structure which affords complete protection of working parts.

3.7.2 Uprights. Uprights shall be of the telescopic type with sufficient strength and rigidity to withstand, without permanent deformation, stresses induced by capacity loads. Adequate means, either by adjustment or replacement, shall be provided for alignment of uprights and to compensate for wear which would produce lateral looseness. Upper and lower limit stops shall be provided to prevent over-travel. The upright lift arrangement shall be designed to control premature elevation of intermediate section(s) during raising or "hanging-up" of these section(s) during lowering.

3.7.2.1 Forks. Steel forks shall be provided which shall be capable of withstanding, without permanent deformation, a static test load of 300 percent of rated load as well as stresses induced by tests specified in section 4.

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Forks shall be attached in a manner which prevents lateral slippage or permanent deformation of lift carriage components. Forks shall be adjustable laterally without the need for hand tools. Fork thickness shall not exceed .05 inch (12mm) at the tip and the end taper shall be not less than 14 inches (356 mm) long. Taper shall be on the underside of the fork. Minimum fork width shall be 4 inches (102 mm); fork length shall be 36 inches (814 mm). Maximum fork spread, forks together shall not exceed 18 inches (457 mm).

3.7.3 Wheels. Trucks shall be equipped with one or two drive wheels and two load wheels. Drive wheel(s) shall be of the flat face or demountable type made from steel or cast iron mounted on ball or tapered roller bearings providing instant travel reversal without scrubbing action. Readily accessible grease fittings shall be furnished to facilitate lubrication in accordance with good automotive practice where pre-lubricated sealed bearings are not provided.

3.7.3.1 Drive wheel tires. The drive wheel tire(s) shall be polyurethane or industrial solid rubber smooth low profile type or cushion type, molded onto a steel band, with non-directional tread. Tires for flat face wheels shall be for pressing onto a flat face wheel. Tires shall be new and unused. Tire(s) may be static discharge type to meet the requirement of paragraph 3.5.5.2.

3.7.3.2 Load wheels. Trucks shall be equipped with polyurethane load wheels mounted on ball or tapered roller bearings. Load wheel ratings shall be sufficient to support maximum loads and operate smoothly without binding. Load wheels may have static discharge type tires to meet the requirement of paragraph 3.5.5.2.

3.7.4 Steering. The truck shall be steered through the drive wheel(s) by a steering handle equipped with hand grips. The diameter of the hand grips shall be between 1.25 (35 mm) and 1.75 (44 mm) inches. The truck shall turn in the same direction as the handle when in forward motion. The forward end of the truck shall be considered the end containing the control unit. The steering handle shall turn the drive wheel assembly at least 90 degrees in each direction from the neutral position. The drive wheel assembly shall be mounted on ball or tapered roller bearings or combination of both. Provision shall be made for precision adjustment of adjustable type bearings where used to compensate for wear and for lubrication of all friction points by accessible grease fittings as specified in 3.4.3 or by a lubricant reservoir. When provided, the lubricant reservoir shall be easily accessible for filling or checking of lubricant level.

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3.7.5 Brakes. Brakes shall be mechanical external contracting or internal expanding or disc type mounted on the traction motor shaft, countershaft, or drive wheel(s). Brake linings shall be of the conventional type, non-asbestos, cement bonded or riveted to the brake shoe or bolted or riveted to the carrier plate. The brake system shall be interlocked with a relay switch to cause disconnection of the electrical circuit from the traction motor when the brakes are applied. The brakes shall be applied automatically when the steering handle is within 10 degrees of the true vertical or horizontal positions. The steering handle shall return from an operating position to a vertical braking position when released. The force required to hold the steering handle in an operating position shall not exceed 10 pounds(4.5 kilograms). Brakes shall be adjustable to compensate for wear and shall be capable of retaining the truck with rated load on a 10-percent grade.

3.7.6 Truck housing. All functioning components of the truck, except steering handle, wheels, mast and fork assembly shall be enclosed in a housing. The housing shall be designed to permit easy access to all working parts requiring inspection, servicing and maintenence. The height and width of the housing shall be such as to provide unobstructed view of fork tips when the operator is in the operating position.

3.7.7 Reaching and tiering features. When specified (see 6.2) the truck shall be furnished with an hydraulically actuated reach mechanism which will permit extension of the fork carriage to a point where the front vertical face is at least 0.75-inch (19mm) in front of the forward edge of the load wheels, tires or outriggers to permit handling of loads at floor level that are off center laterally or would otherwise conflict with the outriggers or load wheels. The reach mechanism structure shall be capable of meeting the performance requirements specified herein and shall be sufficiently strong and well braced to prevent racking or twisting that might result in permanent distortion of any component part. The design must be particularly well suited to withstand stresses which result from bumping of one fork on the corner of the load or against an obstruction. Antifriction bearings, bronze or steel bushings shall be provided for the rollers on the pantograph type extension and the extensible mast reach mechanism.

3.8. Controls and instrumentation. Where applicable, the provision of UL 583 shall apply. Electric controls shall be constructed to be moisture and weather resistant to prevent entry of moisture and rain when operated or stored out of doors under all weather conditions. All controls shall be identified and provided with permanent position markings.

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3.8.1 Directional speed controls. Directional and speed controls (rotating handles, buttons, levers, or switches) shall be mounted on the steering handle and shall be conveniently located for right or left hand operation. Means shall be provided to prevent simultaneous actuation of forward and reverse controls. Controls shall be arranged so that forward movement will cause forward motion of truck and rearward movement will cause rearward motion. Trigger speed controls are not acceptable. If buttons or switches are used, the one located nearest the operator shall, when actuated, cause forward motion of the truck and vice versa. The controls shall be designed so that they will return to neutral position when released. The operating head of the control handle shall be designed to protect the operator's hands from injury due to possible contact of the handle with storage racks, pallets, unit loads, walls, or any other objects. This design shall allow sufficient room for easy freedom of movement of the operator's hands and shall be an integral part of the handle. A safety reverse control shall be incorporated in the extreme end of the handle to reverse the direction of the equipment in the slowest speed, should the handle come in contact with the operator.

3.8.2 Lifting and lowering controls. The lifting and lowering control(s) shall be lever or button actuated type. If buttons are used for both lifting and lowering, the one located nearest the operator or the upper, if arranged vertically, shall, when actuated, raise the forks. The lift control shall be arranged so that the forks will remain at any position reached when the control is released. Deactivated controls shall return to the neutral position.

3.8.3 Lift, reach and tilt controls. When a reaching and tiering truck is specified (see 6.2), lift, reach and tilt controls shall be self-centering. Rearward or upward motion of lift, reach and tilt control levers shall elevate, retract or tilt load rearward. Modulation of the speed of lift, or lowering, reach and tilt shall be readily controllable. Push button tilt control on reach control handle is acceptable.

3.8.3.1 Extension-retraction control. The forks shall be extended by movement of the control from the center position to the forward position. Retraction of the forks shall be accomplished by movement of the control from the center position to the rearward position. Means shall be provided to prevent jerky stopping or starting of forks at the end of the extension and retraction movements.

3.8.4 Instruments. All gages and meters shall be products of manufacturers regularly engaged in producing these types of instruments.

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They shall be flush-mounted in a location convenient to and plainly visible to an observer standing on the floor.

3.8.4.1 Hour meter. An electric hour meter, non-braked, flush-mounted, direct reading type, conforming to MIL-M-3971 type I, Grade A and UL 583 shall be furnished. The meter shall register the total number of hours of operation of the traction and pump motors. Quartz hour meters are also acceptable.

3.8.5 Horn. Trucks shall be equipped with an electric horn with push button mounted in the steering handle. The horn button assembly and electrical wiring shall be constructed to be moisture and weather resistant.

3.9 Floodlight. Each truck shall be equipped with one seal beam floodlight of not less than 35 watts, mounted on the left side of the mast. Directional focus of this light shall be vertically and horizontally hand adjusted without the use of hand tools. Vertical adjustment of this light shall be a minimum of 60 degrees above and 30 degrees below horizontal plane. Horizontal adjustment shall be a minimum of 200 degrees rotation. When specified (See 6.2), trucks shall be equipped with an identical floodlight mounted on the right side of the mast. When specified (See 6.2), each truck shall be equipped with four floodlights, two in front and two in rear, of which at least one front light and one rear light shall be hand adjustable. Adequate protection by position or guard shall be provided to prevent damage to the light under operating conditions.

3.10 Performance.

3.10.1 Speed. While traveling on a level surface, trucks shall be capable of forward and backward speeds of not less than 2 miles per hour (3.2 kilometers per hour) with rated load and not less than 2.5 miles per hour (4 kilometers per hour) without load.

3.10.2 Lifting speed. Lift speed with rated load (ambient temperature 65 degrees to 75 degrees F (18 degrees to 24 degrees C)) shall be not less than 12 feet per minute (4 meters per minute) from floor to maximum fork height and not less than 25 feet per minute (8 meters per minute) without load under the same conditions.

3.10.3 Lowering speed. Lowering speed of the unloaded forks (ambient temperature 65 degrees F (18 degrees C) to 75 degrees F (24 degrees C)) shall be not less than 30 ft per minute (9 meters per minute).

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The hydraulic control system shall be damped, metered, or manually regulated to reduce shock and prevent damage to the trucks when rated load is lowered at maximum speed and stopped suddenly.

3.10.4 Turning radius. Trucks shall be capable of turning 360 degrees in either direction within a circle the radius of which is not greater than 70 inches (1778 mm). All parts of the truck structure shall remain within the applicable circle during the test.

3.10.5 Slope ascension. Trucks shall be capable of ascending a 10-percent grade on a dry concrete surface when carrying rated load. In addition, trucks shall be able to accelerate from a dead stop on this slope carrying rated load.

3.10.6 Tilt. When reaching and tiering feature is specified (see 6.2) unloaded truck forks or fork carriage shall have forward tilt not less than 2.5 degrees, and backward tilt of not less than 3.5 degrees, measured from the true vertical position of the mast.

3.10.7 Upright height. Collapsed mast height of unloaded truck, measured from the ground to the top of the uprights in a vertical position shall not exceed 83 inches (2108 millimeters) or as specified for size 4 truck (see 6.2).

3.10.8 Fork height. Minimum fork height of loaded truck, measured from the ground to the top face of the forks with uprights vertical, shall be 130 inches (3300 millimeters) or as specified for size 4 truck (see 6.2).

3.10.9 Free lift height. Height of forks without increasing specified collapsed mast height shall be as specified in 3.10.9.1 or 3.10.9.2, whichever is required (see 6.2).

3.10.9.1 Limited free lift. Height of forks shall be not less than 6 inches (150 mm) with rated load on forks.

3.10.9.2 Full free lift. Height of forks shall be not less than 20 inches (510mm) or as specified for size 4 truck (see 6.2) with rated load on forks.

3.10.10 Underclearance. The truck shall have sufficient underclearance to permit operation from one horizontal plane to another, up or down a 10 percent slope with rated load on the forks.

3.10.11 Hydraulic system. The hydraulic system shall show no permanent deformation or failure or leakage when the truck is

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subjected to an ultimate test load of 300 percent of rated load. This requirement does not include the pump. The pressure relief valve(s) shall be set to open when the pressure equals 110 percent of the maximum pressure needed to lift and tilt rated load.

3.10.12 Lift assembly. The lift assembly shall be capable of completing 200 endurance cycles as described in 4.5.2.10 with no signs of damage to the hydraulic lines or appreciable wear. "Appreciable wear" is defined as any apparent inability to lift and hold rated load with not more than 0.5 inch (12mm) drift per minute.

3.10.13 Brake. The brake shall be capable of stopping the truck carrying rated load from maximum speed in accordance with applicable requirements of ANSI B56.1 when tested in accordance with 4.5.2.11.

3.10.14 Stability. The truck shall conform to the stability requirements of ANSI B56.1 when tested as specified in accordance with 4.5.2.13.

3.10.15 Steering. The force required at the steering handle (with the hand-grip of the steering handle located 40 inches, (1016 mm) plus or minus 1 inch (25mm) above the floor surface) to turn the drive wheel(s) in either direction when the truck is at a static position with and without rated load shall not exceed 70 pounds (310 Newtons). This requirement shall be tested as specified in 4.5.2.3.2.

3.10.16 Extension and retraction. When reaching and tiering truck is specified (see 6.2), the truck shall be capable of extending rated load (ambient temperature between 65 degrees and 85 degrees F. (18 degrees and 29 degrees C.)), at any height; a horizontal distance not less than the distance from the forward face of the forks (retracted) to at least 0.75 inches (19mm) forward of outermost end of the outriggers or load wheels, whichever is further, in not more than 5 seconds.

3.10.16.1 Retraction overload. Forks, mast assembly and related structures shall show no permanent deformation or failure when subjected to a retraction overload of 300 percent of rated load when tested as specified in 4.5.2.19.1.

3.10.16.2 Extension overload. Forks, mast assembly and related structures shall show no permanent deformation or failure when subjected to an extension overload of 200 percent of rated load when tested as specified in 4.5.2.19.2.

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3.11 Interchangeability. Corresponding units and replaceable assemblies, sub-assemblies, and parts having the same part number shall be physically and functionally interchangeable as complete items without modification. When dimensions, ratings, and characteristics, are not specified, the manufacturer's design limits shall be used to determine compliance.

3.12 Finish.

3.12.1 Treatment and painting. Exterior surfaces of the truck shall be thoroughly cleaned, dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not less than one coat of primer and two coats of synthetic enamel. Primer and finish coats shall be applied in accordance with normal commercial practices to assure complete coverage and durability of the finish. The finish color shall be yellow conforming to FED-STD-595, color number 13538 except mast may be manufacturer's standard matte black color. The finish coat, when dry, shall be a smooth, even surface, free from runs, sags, peels, chips, blisters, and areas of no film. Surfaces of components and assemblies of the truck, not normally painted with a finish coat shall be cleaned and prepared in accordance with standard commercial practice for the forklift truck industry. Use of paint containing lead or chromates is prohibited.

3.12.2 Plating. Fastening devices such as cotter pins, washers, nuts and bolts required to fabricate the truck except those in contact with oils in reservoirs and those inside corrosion resistant components shall be zinc plated, cadmium plated or made of corrosion resisting material. Zinc plating shall be in accordance with ASTM B633, type I, SC 1 and cadmium plating shall be in accordance with QQ-P-416, type I, class 3.

3.13 Electromagnetic interference control. Unless otherwise specified (see 6.2), the trucks shall be equipped for control of electromagnetic interference in accordance with SAE J551C.

3.14 Rain. When specified (see 6.2), the truck shall be capable of starting and performing all normal operations as specified herein when exposed to a simulated rainfall up to 4 inches (102 mm) per hour with a 30 mile-per-hour (48 kilometers per hour) wind when tested as specified in 4.5.2.18.

3.15 Identification marking. The manufacturer shall provide and install a metal nameplate according to MIL-STD-130 in a visible and safe location on the truck with the following information indented or embossed or otherwise permanently marked on the plate:

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TRUCK, LIFT, FORK, TELESCOPING MAST,ELECTRIC
SELF PROPELLED (WALKIE)

CAPACITY .....pounds (kilograms) at 24" (610 mm) load center
NAME OF MANUFACTURER.....
SERIAL NUMBER.....
MANUFACTURER MODEL NUMBER.....REGISTRATION NUMBER.....
DELIVERY DATE.....(month and year)
WARRANTY EXPIRATION DATE.....(month and year)
CONTRACT NUMBER.....
INSPECTED.....(name of inspection office)
SERVICE WEIGHT.....pounds (less battery)

The plate shall conform to MIL-P-514, type III, composition C, (GG-P-455, type I, grade A, class 1).

3.15.1 Instruction plates. Each truck shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in loading, operating, or servicing the truck. Instruction plates shall conform to MIL-P-514, type III, composition C, (GG-P-455 of type I, grade A, Class 1) material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location. Vinyl sheet plates with self adhesive backing may also be used.

3.15.2 Safety rating. Each truck shall be equipped with a safety designation plate conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, Class 1). As a minimum, the plate shall indicate truck manufacturer's name, truck's complete model number, safety designation (type E, EE), independent testing laboratory's name, independent laboratory's registration or index number assigned to the inspected truck and date and location of inspection.

3.15.3 Battery plate. Each truck shall be equipped with a battery information plate attached to the truck conforming to MIL-P-514, type III, composition C (GG-P-455,type I, grade A, class 1) material. As a minimum, the plate shall indicate battery weight, dimensions, ampere hour capacity at 6-hour rate, voltage, number of cells, number of plates, cable lead length and position.

3.15.4 Vehicle marking. The lifting capacity of the truck shall be painted in block numbers three inches high on each side of the mast. When the mast is painted black, markings placed on the mast shall be in white enamel.

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The assigned registration number for each truck shall be painted in block numbers three inches high on each side of the truck. Durable black enamel markers indicating the safety designation of the truck shall be applied on each side and on the rear of the truck in a visible but protected location. These markers shall be in accordance with UL 583. In addition, supplementary marking and load handling symbols as specified in ANSI B56.11.3 are also required. "No riders" warning shall be painted in block numbers two inches high on the rear of the mast. Block numbers shall be painted in black enamel. The words "Painted with lead free and chromate free paint on (MM/YY) by (Contractor Name)." shall be painted in block letters 1-1/2 inch (38mm) high on each side of the truck in a visible but protected location.

3.16 Registration number. The assigned registration number for each truck shall be painted by the contractor in black enamel in block numbers 3 inches (76mm) high on each side of the truck chassis.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced truck inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

(c) Inspection of preparation for delivery (see 4.7)

4.3 First-produced truck inspection. Trucks furnished in accordance with 3.2 shall be examined for the defect marked "X" in column 1 Table I and subjected to tests marked X in column 1 of Table II. Acceptance of a first-produced truck shall not exclude the remaining trucks from the quality conformance inspection requirements. Failure of a first-produced truck to meet any examination and test specified shall be cause for rejection of the first-produced truck.

4.4 Quality conformance inspection.

4.4.1 Examination. Each truck shall be inspected for the defects marked "X" in column 2 of Table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each truck shall be subjected to tests marked X in column 2 of Table II. Failure to meet any test requirement shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Truck(s) shall be examined for defects as specified in 4.3 or 4.4.1 as applicable.

4.5.2 Tests. Tests to determine compliance with the specified requirements shall be conducted as outlined herein.

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TABLE I Examination Schedule

First Produced	Inspection	Quality	Conformance	Defects	Requirement Paragraphh
1		2		3	4
X		X			
X		X		Components missing or not as specified	Section 3
X		_		Materials not as specified	3.3
X		-		Safety requirements not as specified-type E or Type EE truck	3.4.1
X		-		Threads not in accordance with Handbook H28	3.4.5
X		-		Overload devices not provided with each circuit	3.5.3
X		-		Thermal relays for resistor housings not installed	3.5.5
X		-		Means of identification for conductors not provided	3.5.2
X		X		Two (2) replaceable static discharge devices not supplied	3.5.5.2
X		-		Vehicle not designed for 24 volt D.C. operation	3.6.1
X		-		Thermal relays for motors not installed	3.6.2
X		-		Contactors for speed controller not easily replaceable	3.6.4
X		-		Battery connectors not as specified	3.6.5.1
X		X		Ventilating holes for battery compartment or well not screened or louvered	3.6.5.3

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TABLE I Examination Schedule

First Produced	Inspection	Quality	Conformance	Defects	Requirement Paragraphh
1		2		3	4
X		-		Drive assembly not as specified	3.6.6
X		-		Hydraulic system not as specified, does not function, or leaks	3.6.7
X		-		Contamination present in hydraulic reservoir return, and suction line	3.6.7
X		-		Piston rod not as specified	3.6.7.3
X		-		Frame depth does not protect working parts	3.7.1
X		-		Fork dimensions not as specified	3.7.2.1
X		-		Ball or tapered roller bearings not provided	3.7.3
X		X		Accessible grease fittings not provided for all rotating parts	3.7.3
X		-		Tires not as specified	3.7.3.1
X		-		Load wheels not made of polyurethane	3.7.3.2
X		-		Steering system not as specified, does not function, or leaks	3.7.4
X		-		Brakes not as specified or do not function	3.7.5

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TABLE I Examination Schedule

First Produced	In-process Inspection	Quality Conformance	Defects	Requirement Paragraphh
1		2	3	4
X		-	Truck body does not completely cover drive unit assembly, steering mechanism and speed controller when these components are not housed in individual containers	3.7.6
X		-	Easy access to all working parts is not provided	3.7.6
X		-	Controls and instrumentation not as specified or do not function correctly	3.8
X		-	All controls are not provided with heavy moisture and weather resistant markings	3.8
X		X	Directional speed controls and push button requirements do not function as specified	3.8.1
X		-	Lifting and lowering controls are not as specified	3.8.2
X		X	Extension and retraction controls not as specified	3.8.3.1
X		X	Instruments not located as specified or do not function correctly	3.8.4
X		X	Treatment and painting not as specified	3.12.1
X		-	Electromagntic interference not as specified	3.13
X		X	Nameplate and special marking incorrect or not as specified	3.15,

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Table II. Test Schedule

First Prod.	Qual. Conform.	Post Test	Test	Test Para.	Requir. Para.
1	2	3			
X	X	X	Speed	4.5.2.1	3.10.1
X	X	X	Lifting and lowering speed	4.5.2.2	3.10.2 3.10.3
X	X	X	Turning radius	4.5.2.3.1	3.10.4
X		X	Steering force	4.5.2.3.2	3.10.15
X		X	Steering handle holding force	4.5.2.3.3	3.7.5
X		X	Upright height	4.5.2.4	3.10.7
X		X	Fork tilt	4.5.2.5	3.10.6
X		X	Fork height	4.5.2.6	3.10.8
X		X	Free lift height	4.5.2.7	3.10.9.1 3.10.9.2
X	X	X	Underclearance	4.5.2.8	3.10.10
X			Overload	4.5.2.9	3.7.2.1 3.10.11
X			Endurance	4.5.2.10	3.10.12
X	X	X	Brakes	4.5.2.11	3.10.5
X	X	X	Slope ascension	4.5.2.12	3.10.5
X			Stability	4.5.2.13	3.10.14
X			Electromagnetic interference	4.5.2.14	3.13
X			Operating temperature	4.5.2.15	3.5.5, 3.6.2
X			Thermal relay	4.5.2.16	
X			Thermal relay temperature	4.5.2.16.1	3.5.5.1, 3.6.2, 3.6.7
X			Drive unit	4.5.2.17	3.6.6
			Rain	4.5.2.18	
X	X	X	Extension and retraction	4.5.2.19	3.10.16
X			Retraction overload	4.5.2.19.1	3.10.16.1
X			Extension overload	4.5.2.19.2	3.10.16.2

4.5.2.1 Speed. The truck, with rated load, shall be accelerated on a dry, smooth level surface through a 50-foot (15 m) to 75-foot (23 m) approach and then timed through a measured interval. Three runs shall be made in both the forward and the reverse directions in high speed. This test shall be repeated without load. Record data and calculate average speeds. Noncompliance with 3.10.1 shall constitute failure of this test and shall be cause for rejection.

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4.5.2.2 Lifting and lowering speeds. Rated load shall be placed on the forks with the truck on a level surface. The lift control shall be activated and the load lifted to maximum lift height. The time in seconds shall be taken using a timing device from the instant forks begin to rise to the end of full lift height. This test shall be repeated at least 5 times and all data tabulated and averaged. The average time required to raise rated load to maximum lift height shall not exceed the limits specified in 3.10.2. The lower control shall be actuated and deactivated to determine compliance with 3.10.3 with no load on the forks. Noncompliance with 3.10.2 and 3.10.3 shall constitute failure of this test and shall be cause for rejection.

4.5.2.3 Steering.

4.5.2.3.1 Turning radius. The truck with rated load equally distributed on the forks or pallet shall be placed on dry level concrete surface. The truck shall be operated to travel in a circular path and observed. Noncompliance with 3.10.4 shall constitute failure of this test and shall be cause for rejection.

4.5.2.3.2 Steering force. The truck, with rated load equally distributed on the forks or pallet shall be placed on a smooth, dry, level concrete surface, with the steering handle grip(s) 40 inches (1016 mm), plus or minus 1-inch (25 mm), above the concrete surface. A tension dynamometer shall be attached at the hand grip. For this test the steering handle holding device shall be made inoperative. A force shall be applied to the tension dynamometer tangent to the arc through which the steering handle moves (horizontal plane). The steering handle shall be made to turn in each direction from the neutral position through an arc of 80 degrees, minimum. Three readings in each direction shall be taken. Record data and calculate average force in each direction to determine. Noncompliance with 3.10.15 shall constitute failure of this test and shall be cause for rejection.

4.5.2.3.3 Steering handle holding force. A tension dynamometer shall be attached at the hand grip. A force shall be applied to the tension dynamometer tangent to the arc through which the steering handle moves (vertical plane). The steering handle shall be rotated about its pivot point (vertical plane) from its uppermost position to its lowest position. The maximum force to hold the steering handle in any position on the arc shall not exceed the force specified in 3.7.5. Noncompliance with 3.7.5 shall constitute failure of this test and shall be cause for rejection.

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4.5.2.4 Upright Height. The truck shall be placed on a level surface with unloaded forks in the lowest position. Measure in inches the vertical distance from the floor to the uppermost projection of the mast assembly. Noncompliance with 3.10.7 shall constitute failure of this test and shall be cause for rejection.

4.5.2.5 Upright tilt. The loaded truck, with rated load in carrying position shall be placed on a solid surface and leveled by use of a liquid level or equivalent placed on the frame structure as near to the lift assembly as possible. A reference line, vertical to the fork surface, shall be located on or to the side of the rated load. The load shall then be raised to one foot above the surface and tilted forward and backward. The angle between the plumb line and the reference line vertical to the fork surface shall be recorded. A clinometer may also be used to measure tilt angle. Noncompliance with 3.10.6 shall constitute failure of this test and shall be cause for rejection.

4.5.2.6 Fork height. Place truck, with rated load equally distributed on the forks or pallet, on a level concrete surface. Elevate load to the maximum lift height and measure in inches (millimeters), the distance from the floor to the top face of the forks to determine compliance with 3.10.8. Noncompliance with 3.10.8 shall constitute failure of this test and shall be cause for rejection.

4.5.2.7 Free lift height. Place the truck on a level surface with rated load equally distributed on the forks or pallet. Elevate forks to maximum height without exceeding the specified collapsed mast height. Measure the distance from the floor to the top face of the forks. Noncompliance with 3.10.9.1 or 3.10.9.2 when specified (see 6.2) shall constitute failure of this test and shall be cause for rejection.

4.5.2.8 Underclearance. The truck shall be tested to determine compliance with 3.10.10. The truck, with rated load on the forks, shall be operated from one horizontal plane up a 10-percent ramp to another horizontal plane. Observe whether all parts of the truck clear the surface. Noncompliance with 3.10.10 shall constitute failure of this test and shall be cause for rejection.

4.5.2.9 Overload test. Place truck on a level surface and elevate empty forks to a minimum of 30 inches (762 mm) lift height. The truck shall be secured to avoid tipping during the test. With the pressure relief valve inoperative, place a static test load equal to 300 percent of rated load on the forks at rated load center. Retain the load for a period of 10 minutes and remove load. The truck shall be inspected to determine compliance with 3.7.2.1 and 3.10.11.

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Noncompliance with 3.7.2.1 or 3.10.11 shall constitute failure of this test and shall be cause for rejection.

4.5.2.10 Endurance test. The truck shall be subjected to 200 cycles of the endurance test. One cycle of the endurance test shall be as outlined herein. The truck shall be required to lift rated load to maximum height and be propelled clockwise in accordance with 3.10.4 through 360 degrees after which the brakes shall be applied. The load shall then be lowered, raised again, and the truck propelled 360 degrees in a counter-clockwise direction and brought to a complete stop. A five-minute rest period may be allowed between each 25 cycles completed. The battery may be changed as required during the test. At the end of 200 cycles, the rated load shall be lifted to maximum lift height and held for a period of 5 minutes. Drift of the load shall not exceed one-half inch (12mm) per minute. After completion of the above tests, the hydraulic system shall be inspected to determine compliance with 3.10.12. Noncompliance with 3.10.12 shall constitute failure of this test and shall be cause for rejection.

4.5.2.11 Brakes. Place rated load on forks and bind to forks. Elevate forks to maximum full free lift height and accelerate truck in a forward direction maximum speed. The truck shall be stopped quickly and smoothly without grabbing. Measure distance required to stop truck after the brakes are applied. Noncompliance with 3.10.13 shall constitute failure of this test and shall be cause for rejection.

4.5.2.12 Slope ascension. The truck with rated load shall be accelerated along a horizontal surface and onto a 10 percent ramp and brought to a dead stop. The truck shall then accelerate up the ramp and be observed. Noncompliance with 3.10.5 shall constitute failure of this test and shall be cause for rejection.

4.5.2.13 Lateral stability. Test the truck in accordance with ANSI B56.1. Noncompliance with 3.10.14 shall constitute failure of the test and shall be cause for rejection.

4.5.2.14 Electromagnetic interference. Unless otherwise specified, the test to determine compliance with 3.13 shall be performed in accordance with the requirements of SAE J551C. Failure to meet requirements shall be cause for rejection of the truck.

4.5.2.15 Operating temperature test. The type EE truck shall be operated at low speed over a level surfaced test course. The test course and method of negotiating the course shall be in accordance with figure 1.

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The truck shall be provided with sufficient heat sensing devices located at critical heat areas on the outside surfaces of the drive motor and resistor housing, and a temperature recording instrument shall continuously record these temperatures during the test run. This test shall be run with continuous operation of truck until temperatures being recorded have stabilized. Temperature shall be considered as being stabilized when the temperature being recorded remains constant over a minimum period of one hour during continuous operation of the truck. Failure to meet requirements of paragraphs 3.5.5 and 3.6.2 shall be cause for rejection.

4.5.2.16 Thermal relay test (for type EE trucks only). Disconnnect the lead to one side of the relay. Activate the control to apply power to the affected component. Reconnect the lead. Repeat the test for each relay. Ability to apply power to the affected circuit with the relay lead disconnected shall constitute failure and shall be cause for rejection.

4.5.2.16.1 Thermal relay temperature test. Representative samples of relays shall be placed in an environment in which the temperatures can be controlled. Gradually increase temperature until relays open. Observe and record temperature. Nonconformance to 3.5.5.1, 3.6.2 and 3.6.7 shall constitute failure and shall be cause for rejection.

4.5.2.17 Drive unit test. When specified (see 6.2), the drive unit assembly shall be tested in accordance with 3.6.6. The drive wheel(s) shall be locked and the specified torque shall be applied to the output shaft of the drive motor for a period of 5 minutes. The applied torque shall then be removed and the drive unit shall be inspected. Noncompliance with 3.6.6 shall constitute failure of this test and shall be cause for rejection.

4.5.2.18 Rain Test. When specified (see 6.2) the truck shall be subjected to simulated rainfall of not less than 4 inches (102 mm) per hour in accordance with 3.14. Simulated rainfall shall impinge on the truck at an angle of 45 degrees, plus or minus 5 degrees, with a force equivalent to a 30 mile-per-hour (48 kilometers per hour) wind. Rainfall shall disperse uniformly over the surfaces of the truck. The test shall be conducted as follows:

(a) Expose the top and left side of the truck to the simulated rain and wind as described for a period of 5 minutes with the power shut off. With rain intensity continuing, start the truck within 1 second (without removing accumulated water) and activate forward and reverse drive as well as lift and tilt operations.

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(b) Repeat (a) with the top and right exposed.

(c) Repeat (a) with the top and front exposed.

(d) Repeat (a) with the top and rear exposed.

At the conclusion of the test, the forklift shall be examined for any leakage into the electric motor, control boxes and hydraulic tank reservoir. One or more of the following shall constitute failure of this test.

(a) Water leakage into the electrical and hydraulic components described above.

(b) Malfunctioning of any electrical component.

(c) Inability to start and operate truck during this test.

4.5.2.19 Extension and retraction. When specified (see 6.2), raise rated load to maximium lift height in five interval steps. Extend and retract rated load at various heights to the full horizonatal distance. Measure this distance and record the time in seconds from the instant extension or retraction begins to the end of extension or retraction. Noncompliance with 3.10.16 shall constitute failure of this test and shall be cause for rejection.

4.5.2.19.1 Retraction overload. With pressure relief valve inoperative, raise retracted forks to minimum of 30 inches (760mm) lift height and place 300 percent of rated load on forks and retain for ten minutes. Remove load and examine truck for evidence of external leaks or breakage of the hydraulic system as a result of this test. The load support assembly, frame, and related structure shall be examined for permanent deformation or any indication of structural failure. Noncompliance with 3.10.16.1 shall constitute failure of this test and shall be cause for rejection.

4.5.2.19.2 Extension overload. Extend load support assembly to full forward position, with forks raised at least 12 inches (300mm) above the floor and rear of truck counterweighted or tied down. Place 200 percent rated load on the fork assembly and retain for 5 minutes. Remove load and examine the truck for evidence of failures. The pressure relief valve shall be made inoperative for this test. Noncompliance with 3.10.16.2 shall constitute failure of this test and shall be cause for rejection.

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4.6 Post-test inspection. Upon completion of all tests specified in 4.5.2, the truck shall be subjected to tests marked X in column 3 Table II. Nonconformance with applicable requirements shall constitute failure.

4.7 Inspection of preparation for delivery. The trucks shall be examined to determine conformance to the preservation, packaging, packing and marking requirements of MIL-STD-162 or ASTM D 3951 as specified.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging and packing shall be in accordance with MIL-STD-162 or ASTM D 3951 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Trucks described herein are intended for moving cargo in and around warehouses, overhaul and repair shops, loading platforms, aircraft hangars, covered loading docks, truck and trailer-loading, railcar loading and for other applications best performed by powered equipment where permissible floor loadings precludes using heavier equipment. Trucks are intended for operation over paved or other relatively level hard surfaces for short distances.

6.2 Acquisition Requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.

(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documemts referenced (see 2.1).

(c) Type and size required (see 1.2, 3.10.7, 3.10.8).

(d) When first article test is required.

(e) When speed controller shall be solid state type. (See 3.6.4).

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(f) When battery is to be furnished by other than the supplier. (See 3.6.5.1).

(g) Type connector required if other than EC or equal. (see 3.6.5.2).

(h) When drive assembly test is required (see 3.6.6).

(i) When reaching and tiering features are required. (see 3.7.7, 3.8.3, 3.10.6)

(j) When additional floodlights are required (see 3.9).

(k) Free lift requirements (see 3.10.9.1 and 3.10.9.2).

(l) Electromagnetic interference requirements (see 3.13).

(m) When rain resistance is required (see 3.14).

(n) Level of preservation, packaging and packing (see 5.1)

(o) Special marking, if required (see 5.2).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a standard production item from the contractor's current inventory (see 3.2) and the number of items to be tested as specified in 4.3. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Conditions for use of level B preservation. When level B preservation is specified (see 5.1), this level of protection should be reserved for the acquisition of forklift trucks for resupply worldwide under known favorable handling, transportation and storage conditions.

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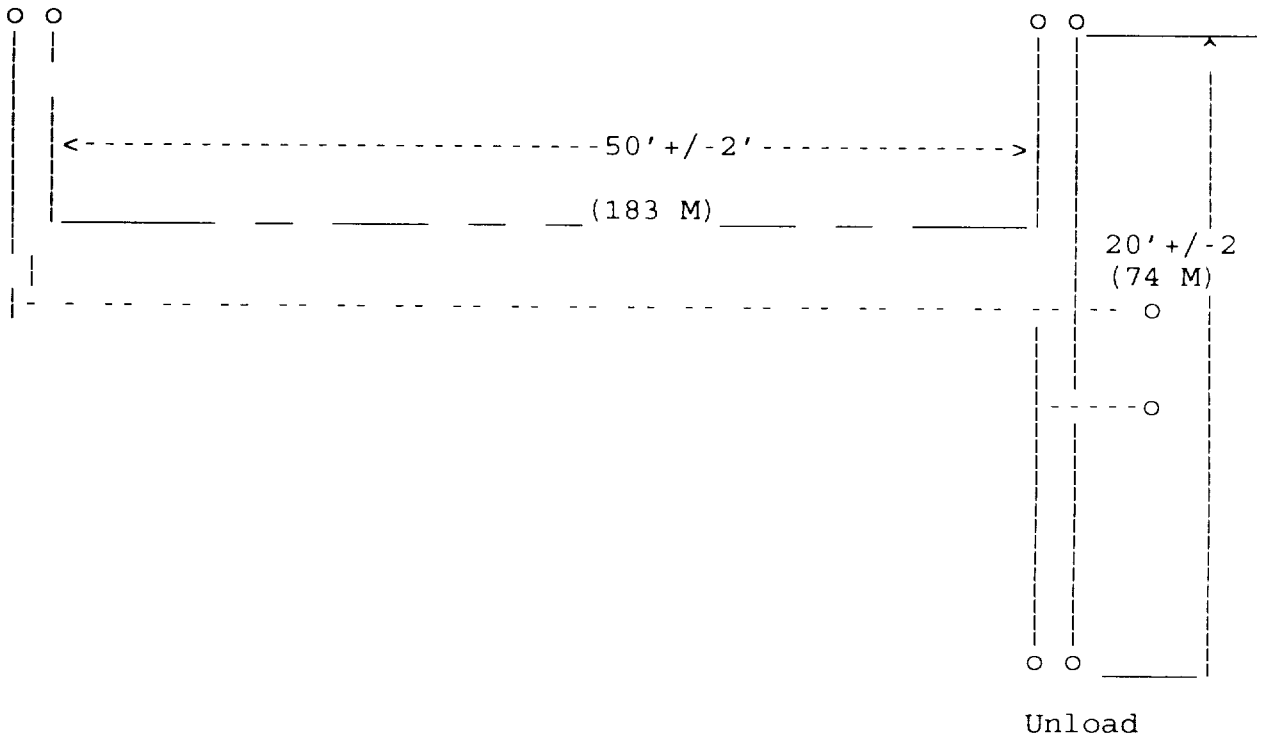
6.5 Subject term (key word) listing.

Equipment, material handling

6.6 Changes from previous issue. Marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Start-pick up rated load

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Reverse, Low Speed, W/Rated Load
Forward, Low Speed, W/Rated Load
Reverse, Low Speed, W/O Rated Load
Forward, Low Speed, W/O Rated Load

Note: On alternate laps, the load should be picked up where it was previously deposited and redeposited wherre it was previously picked up. The load shall be raised to full lift height before it is deposited.

TEST COURSE FOR ELECTRIC (SPARK ENCLOSED) FORK LIFT TRUCK

(MOTOR AND RESISTOR HOUSING HEAT STABILIZATION)

FIGURE 1

Custodians: Preparing Activity

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Custodians:
Army - ME

Navy - SA

Air Force - 99

Review Activities:

Army - AT(AMSA-GDS)

Navy - MC

Air Force - 84

DLA - CS

User Activities:

Navy - OS, MS, AS

DLA -DH

Preparing Activity:
Navy - SA

Project No.

3930-0673

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-21867D	941031

3. DOCUMENT TITLE

TRUCK, LIFT, FORK, TELESCOPINT MAST, ELECTRIC, SELF PROPELLED (WALKIE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
NAVY SHIPS PARTS CONTROL CENTER	(1) Commercial: (717) 790 4024	(2) AUTOVON: 430 4024
c. ADDRESS (Include Zip Code) 5450 CARLISLE PIKE, P.O. BOX 2020 MECHANICSBURG, PA 17055-0788	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

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